# COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 2 : Session 1 : SLO 1 — Functional Units of a Computer
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## Section A — Match the Following

- 1. ALU → a. Executes arithmetic & logic operations
- 2. Control Unit → b. Manages and coordinates all units
- 3. Memory Unit → c. Stores data and instructions
- 4. Input Unit → d. Converts data into computer acceptable form
- 5. Output Unit → e. Displays results to the user

## Section B — Fill in the Blanks

- The ALU unit performs all arithmetic and logical operations in a computer.
- The Control unit manages the coordination of all the units.
- Memory unit stores data, instructions, and intermediate results.
- The Input unit accepts data and instructions from the user.
- The Output unit converts machine results into a form understandable by the user.

## Section C — Activity Task: Block Diagram of Functional Units

- Block diagram: INPUT UNIT → MEMORY UNIT ↔ CPU (CONTROL UNIT + ALU) → OUTPUT UNIT, with the Memory Unit connected bidirectionally to the CPU, and the Control Unit sending control signals to every other unit.
- ALU — (1) Performs all arithmetic operations such as addition, subtraction, multiplication and division. (2) Performs logical operations such as AND, OR, NOT and comparisons used for decision making.
- Control Unit — (1) Fetches instructions from memory, decodes them and generates control signals to coordinate all units. (2) Manages the timing and sequencing of data flow between memory, ALU and I/O units.
- Memory Unit — (1) Stores data, instructions and intermediate results during processing (primary storage: RAM/ROM). (2) Holds final results before they are sent to the output unit.